

Glossary

aerobatics

Spectacular and unusual feats of flying performed for entertainment. The word itself comes from “acrobatICS.”

aerodynamics

The science that analyzes the behavior of air and other gases in motion, and the interaction between air and objects moving through it. It also refers to the aerodynamic properties of a particular object.

aeronautics

Both the science and art of controlled flight, and the science of designing and building aircraft.

afterburner

An arrangement in some military jet aircraft in which extra fuel is burned in the back part of the engine, using air that has already passed through the turbines. It provides extra acceleration but uses a lot of fuel, so it is normally used only on takeoff or in combat situations.

aileron

A movable control surface on the outer part of the trailing edge of an aircraft wing. Ailerons usually work in pairs to alter the relative lift generated by each wing, with one angled downward while its opposite on the other wing is angled upward. In this way ailerons act to control roll, and they also aid in turning movements. In some situations it aids control not to have opposite ailerons moving in an identical extent; a system that allows this is called a differential aileron. *See also* control surface, flaps, lift, roll.

air cooling

Using a flow of air to cool an engine. *See also* liquid cooling.

airframe

All the structural elements that hold an aircraft together, not including engines or other fittings.

air-to-air missile (AAM)

A radar- or heat-guided missile fired at an enemy aircraft or missile from an aircraft.

Alclad

The trade name for a widely used system in which the metal skin of an aircraft is made of a surface layer of corrosion-resistant pure aluminum, which is bonded to stronger aluminum alloys beneath.

altitude

In aircraft contexts, this usually refers to height above sea level, rather than height above the ground below.

amphibian

An aircraft that can operate from both water and land.

angle of attack

The angle at which an airplane’s wing (or the blade of a helicopter or propeller) meets the oncoming air. An airplane usually flies with the front of its wings higher than the back so that oncoming air pushes the wing upward, and the air is deflected downward.

anhedral

An arrangement of an aircraft’s wings in which they are angled downward from the horizontal, so that the tips of the wings are lower than the point at which they are attached to the fuselage. *See also* dihedral.

antiservo tab

A leverlike device attached to some control surfaces that has the effect of increasing the effort required to move that control surface. It has a safety function in making the control seem heavier to the pilot in situations where there might be a danger of forcing the control surface too far. *See also* control surface, servo tab.

artificial horizon (AH)

See attitude indicator.

aspect ratio

The ratio between the length of a wing from fuselage to wing tip and its width from front to back. For example, a long thin glider wing has a high aspect ratio.

attack aircraft

An aircraft specialized for precision low-level attacks on ground targets, often in support of attacking land forces. This distinguishes it from the less precise operation of a conventional bomber. In practice, many military planes have become multipurpose, and combine ground attack roles with other capabilities. Also known as a ground attack aircraft.

attitude indicator (AI)

An instrument in an aircraft’s cockpit that displays the aircraft’s orientation in relation to the ground in terms of both its pitch and roll. It is particularly important in poor visibility conditions. Also known as an artificial horizon. *See also* pitch, roll.

autogyro

An early type of rotorcraft invented in the 1920s. It was created to avoid the problem of stalling that was common in conventional aircraft of the time. The rotor provided lift when turning, but unlike the later helicopter it was not powered, and the aircraft was pulled forward by a conventional propeller.

automatic pilot

An airborne electronic system that automatically stabilizes an aircraft about its three axes, restores it to its original flight path after a disturbance, and can be preset to make the aircraft follow a particular flight path.

bank

To travel with one side of the aircraft higher than the other when turning. In an aircraft, the pilot must combine roll and yaw to achieve this maneuver. *See also* roll, yaw.

barnstormer

In aviation terms, a barnstormer was a pilot that toured around the country, especially in the United States, giving daredevil aerobatic displays. Barnstorming was an event of the post-World War I years, when there were many unemployed pilots. Although on occasion pilots would fly

through buildings, such as barns, the word itself was previously applied to popular touring actors. *See also* aerobatics.

biplane

An airplane with two pairs of wings, one above the other, usually connected by wires and struts to form a cross-braced structure that is light in weight and very stiff. The bracing wires and/or struts induce more drag than a cantilever monoplane wing. *See also* monoplane.

bizjet

Short for “business jet,” any of a variety of usually small jet aircraft developed mainly for the use of private companies. Usually has a capacity of fewer than 20 people.

blimp

The name applied (in Britain at first) to a small, nonrigid airship and later to a tethered barrage balloon used for air defence during World War II.

Borate bomber

A specialized aircraft used to drop fire-retardant chemicals to stop the spread of wildfires. The term comes from a time when borate salts were widely used as fire retardants, although these have now been replaced by other substances as they have an adverse effect on soils.

bore

The diameter of the cylinder of a piston engine. A larger bore provides a larger combustion chamber, and therefore a potentially more powerful engine. *See also* cylinder, piston engine.

bungee

Also termed bungee cord or bungee rope, any of various elasticated or springlike devices used as shock absorbers or to aid movement, for example, in undercarriages.

bushplane

Any airplane used for accessing remote or wilderness areas. A bushplane has to be able to cope with takeoff and landing on short airstrips and rough ground. Features such as having wings positioned high on the fuselage and a taildragger landing gear formation are desirable in such aircraft. *See also* taildragger.

camshaft

The rotating shaft in a piston engine that lifts the engine’s valves in sequence to regulate the introduction of air and removal of exhaust gases. *See also* overhead camshaft, overhead valves, piston engine.

canard

An arrangement in which a pair of small wings or fins is set further forward on the fuselage than the main wings. The term comes from the French word for “duck,” and reflects the similarity in outline of a canard aircraft to that of a flying duck. Canards are sometimes additional to a tail assembly but generally replace it. For stable operation, the arrangement requires sophisticated control systems, but it does offer stall-proof handling.

cantilever

A structure designed to be self-supporting when anchored at only one end. A cantilever wing is one that is supported internally, with no external struts or wires necessary.

carburettor

A device in which fuel is drawn into the combustion chamber of an engine by the suction of the incoming air. In modern engines it has largely been replaced by fuel injection systems. *See also* fuel injection.

center of gravity

Also called center of mass, the single point in any solid object, such as an aircraft, at which gravity can be thought to be acting. Pitch, roll, and yaw all occur about an aircraft’s center of gravity. For an airplane on the ground, the center of gravity has to lie between the main undercarriage wheels and either the nose wheel or the tail wheel (depending on the aircraft design) so that the plane will not topple over. *See also* pitch, roll, yaw.

center of lift

The point within an aircraft (or part of it, such as a wing) where the lifting forces are in balance. The center of lift of a wing needs to be calculated accurately during the design process, so that forces do not become unbalanced during flight.

center of pressure

An imaginary point within an aircraft where all the forces created by the air moving around it can be regarded as being in balance. It is an important concept when designing aircraft so that they will settle back by themselves to a stable position in flight after small adjustments to controls have been made.

centrifugal force

A force generated by rotation that tends to propel objects outward from the center of rotation.

Civil Aviation Authority (CAA)

The public authority regulating civil aviation in the UK. Established in 1972, it has wide responsibilities for licensing flight crew and airports, registering aircraft, and supervising airworthiness regulations. *See also* Federal Aviation Authority.

cockpit

The compartment in which an aircraft’s pilot sits. Cockpits of large aircraft are also called flight decks.

collective control

A control in a helicopter that changes the angle of all of its rotor blades at the same time. This affects the overall lift generated and, therefore, whether the helicopter ascends or descends. *See also* cyclic control.

combustion chamber

Any of the chambers in an engine in which fuel is ignited and burned.

compressor

In a jet engine, an arrangement of rotating, bladed wheels that acts to compress the incoming air before it enters the combustion chambers. It is powered by the engine’s turbines. A compressor either pushes the air backward in a straight line (axial compressor) or radially outward (centrifugal compressor).

control surface

Any of the various movable surfaces on the wings or tail of an airplane that the pilot can adjust to control the craft. They include ailerons, rudder, elevators, flaps, slats, and spoilers. *See also* aileron.